

FINAL EXPLANATION: CORE MATHEMATICS GRADE 10

Teacher Key: Exemplar Answers and Marking Guide

FINAL EXPLANATION: CORE MATHEMATICS GRADE 10 – STATISTICS I (3.1) – FINAL EXPLANATION: THE MAIZE PRICE MYSTERY AT WAKULIMA MARKET

Teacher Key: Exemplar Answers (do not give to students)

INSTRUCTIONS GIVEN TO STUDENTS

Welcome to your Final Explanation! You have spent eight lessons investigating the Maize Price Mystery: a Nairobi family says the price of a 2 kg packet of maize flour at Wakulima Market varies wildly, while their neighbour says it is always about the same. Use the data table below and everything you have learned to answer each section in the space under the prompt. Show every calculation step, write in full sentences, and use words such as data, frequency, mode, mean, median and histogram where they fit. Your answers should always connect back to the driving question: How can we use statistics to tell an honest and useful story about data from our own Kenyan community, and help people make better decisions?

Data: 40 daily prices of a 2 kg packet of maize flour at a Wakulima Market stall over eight weeks.

Price (KSh)	Frequency
98	16
105	12
112	8
120	4

Part 1: From raw chaos to a frequency table

In Lesson 1 the 40 maize prices looked like a chaotic list of numbers that could not settle the argument. (a) Explain why a raw list of data makes it hard to decide whether the family or the neighbour is right. (b) Look at the frequency table above. Check that the frequencies add up to the correct total and say what this total represents. (c) Which price is the mode? What percentage of the 40 days had

(a) In a raw list the numbers jump up and down with no order. Different people notice different numbers, such as one expensive day or a run of cheap days, and reach different conclusions. Nobody can see which prices are common or rare, so the argument cannot be settled fairly. (b) $16 + 12 + 8 + 4 = 40$. The total frequency (Σf) equals the number of daily prices recorded, so no price has been missed or counted twice. (c) The mode is KSh 98 because it has the highest frequency (16). $16 \div 40 \times 100 = 40\%$ of the days. (d) KSh 112 or higher: $8 + 4 = 12$ days. $12 \div 40 \times 100 = 30\%$ of the days. So most days (70%) were at KSh 98 or KSh 105, and expensive days were fewer but not rare.

this price? (d) How many days had a price of KSh 112 or higher? Express this as a percentage of all days.	
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Part 2: Calculating the mean and finding the median

(a) Add a column fx (price $\times$ frequency) to the table and find Σfx. Then calculate the mean price using $\text{mean} = \Sigma fx \div \Sigma f$. (b) Use the frequencies to find the median price. Explain how you decided which prices sit in the middle of the 40 ordered values. (c) Write the mode, median and mean in one line and say whether they are very different from each other.	(a) fx: $98 \times 16 = 1568$; $105 \times 12 = 1260$; $112 \times 8 = 896$; $120 \times 4 = 480$. $\Sigma fx = 1568 + 1260 + 896 + 480 = 4204$. $\text{Mean} = 4204 \div 40 = \text{KSh } 105.10$. (b) With 40 values the median is the average of the 20th and 21st values when the prices are put in order. The first 16 values are KSh 98 (cumulative frequency 16). The next 12 values are KSh 105, so positions 17 to 28 are KSh 105 (cumulative frequency 28). The 20th and 21st values are both KSh 105, so the median is KSh 105. (c) Mode = KSh 98, median = KSh 105, mean = KSh 105.10. They are fairly close, all within about KSh 7 of each other. The mean is slightly above the mode because the few expensive days (KSh 112 and KSh 120) pull it upwards.
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Part 3: What the shape of the data shows

A histogram shows the full shape of the data, not just one number. (a) Look at how the frequencies change as the price goes from KSh 98 up to KSh 120. Describe the shape you would expect a histogram of these prices to have, and name that kind of shape. (b) Explain why this shape helps to explain why the neighbour and the family experienced the market differently. (c) Give one reason why a histogram can tell the story more fully than the mean, mode or median alone.	(a) The frequencies fall as the price rises (16, 12, 8, 4). The tallest bars would be on the left at the lower prices, with a thinning tail towards the higher prices. This is a right-skewed shape. (b) Most days (28 of 40) were at KSh 98 or KSh 105, so a shopper who visits on typical days feels that the price is 'always about the same', as the neighbour said. The tail shows that there are still some expensive days (KSh 112 and KSh 120). A shopper who meets one of these feels that prices jump around, as the family said. (c) A single number such as the mean, mode or median hides how the prices are spread out. A histogram shows the typical prices, the rarer high prices and the overall spread together, so a reader is less likely to be misled.
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Part 4: Which number tells the truth?

At the school canteen, suppose the daily price of mandazi is	(a) The mean would be pulled most. It is calculated by adding every value, so one very high price raises the total and drags the mean up, even though it happened only once. The median depends only on the middle position and the mode only on the most
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about the same each day, except for one day when a sugar shortage in the Rift Valley causes a single very high price of KSh 45. (a) Which of the three measures (mean, median, mode) would be pulled most by that one extreme price, and which would stay stable? Explain why. (b) Now write a short verdict of 3 to 5 sentences answering: 'Who was right in the Maize Price Mystery: the family or the neighbour?' Use at least two of your calculated values from Part 2 and the percentages from Part 1 as evidence.	frequent value, so both stay stable. This is why the mean is sensitive to extreme values and can misrepresent the typical price. (b) Both the family and the neighbour were telling the truth about what they experienced. The mode is KSh 98 and the median is KSh 105, and 70% of the days had a price of KSh 98 or KSh 105, so the neighbour, who usually shops on typical days, found prices 'always about the same'. However, 30% of the days had prices of KSh 112 or KSh 120, and the mean of KSh 105.10 is slightly above the mode because of these days. The family, who met some of these expensive days, truly experienced variation. Statistics settles the argument not by saying one person is wrong, but by showing that different measures and the shape of the data capture different parts of the same reality.
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Part 5: Honest statistics for community decisions

A Nairobi family wants to budget for buying one 2 kg packet of maize flour each week at Wakulima Market. (a) Using your results, which price or prices would you advise them to plan around, and which measure(s) did you use? (b) Explain what they should still prepare for. (c) Why is it important to report data and choose a statistic honestly when people use it to make decisions? Give one example from this unit of how statistics could mislead.	(a) I would advise the family to plan for about KSh 98 to KSh 105 per packet. The mode (KSh 98) and the median (KSh 105) describe a typical day, and the mean of KSh 105.10 gives almost the same budget figure. (b) They should also prepare for the occasional expensive day. 30% of the days were KSh 112 or KSh 120, and KSh 120 occurred 4 times in 40 days (10%). They could keep a little extra money aside for those days. (c) Families, traders and leaders make real decisions, such as how much to budget or when to buy, from the numbers they are shown. If only one statistic is reported, or the data is recorded carelessly, people can be misled. For example, a mean inflated by one extreme price, such as a KSh 45 mandazi caused by a sugar shortage, would make a normal day look expensive. Honest statistics means recording every value accurately, choosing the measure that fits the question, and showing the full picture with a table or histogram. This is integrity in action.
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Criterion	Excellent (4)	Proficient (3)	Developing (1–2)
Accurate calculation and use of the frequency table (total, mode,	All calculations are correct and fully shown, including the $\Sigma fx =$	Most calculations are correct with steps shown; one small slip, such as a	Several calculations are missing or incorrect, or the median and mean are

percentages, Σfx , mean, median)	4204, mean = KSh 105.10, and the median found from the cumulative frequencies.	percentage or the median position, does not stop the answer making sense.	confused or no steps are shown.
Understanding of statistical tools and the shape of the data	Clearly explains why raw data cannot settle the argument and why a frequency table and histogram can, and correctly describes the right-skewed shape from the frequencies.	Explains the purpose of tables and histograms and describes the shape, but with limited detail or links to the data.	Gives vague or incorrect statements about tables and graphs, or cannot describe the shape from the frequencies.
Reasoning about mean, median and mode (including the effect of extreme values)	Correctly explains that the mean is pulled by an extreme value while the median and mode stay stable, and explains why each measure suits a different question.	Identifies that the mean is affected by extreme values but the explanation of why is incomplete.	Does not distinguish the measures or gives incorrect reasons.
Evidence-based verdict and community decision-making	Gives a clear verdict that both the family and the neighbour are right, supported by at least two specific values from the data, and offers a sensible budgeting recommendation with a note on honest reporting.	Gives a reasonable verdict and recommendation with some evidence from the data.	Gives an opinion with little or no evidence from the data, or the recommendation does not match the results.

ATTRIBUTION AND LICENSING

Curriculum Alignment: Structured in pedagogical alignment with the competence-based curriculum format and learning outcomes established by the Kenya Institute of Curriculum Development (KICD). This material is independently developed and is not officially endorsed or published by KICD.

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Production & Layout: Educational design and curriculum sequencing by SeaVuria. Technical layout, structural formatting, and digital optimization/implementation executed in partnership with ARES Education (registered 501(c)(3) in the United States and Canada (as Afretech Aid) and registered NGO in Kenya), with lesson drafting and compilation assisted by AI (Claude, Anthropic).

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